

Mohammed Farah

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EDUCATION

Duke University

Durham, NC

B.S. in Engineering – Electrical and Computer Engineering & Cultural Anthropology (Double Major)

May 2027

Relevant Coursework: Intro to Microelectronic Devices (oscilloscope, function generator, multimeter, DC power supply)

EXPERIENCE

Baini Technologies (HemoSave Device)

January 2024 – Present

Co-Founder / Product Design & Research

Durham, NC

- Addressed critical gaps in postpartum hemorrhage monitoring (70,000+ maternal deaths annually) by conducting user research with 20+ medical professionals, translating clinical needs into detailed technical product specifications and validation metrics.
- Evaluated competing blood-loss measurement methods and built a competitive analysis that directly shaped the device's sub-\$500 cost target and <2 min speed requirement, positioning HemoSave against visual estimation (50% accurate), gravimetric (10+ min), and premium devices (\$2K+).
- Defined product requirements and testing protocols achieving 95%+ correlation accuracy, matching premium-device performance at <\$500 cost and <2 min measurement time while balancing technical feasibility, safety, and manufacturability.
- Awarded **1st place** at the Rice 360 Global Health Design Competition among 20+ international teams, recognized for technical rigor and real-world clinical impact.

HDTP

July – August 2025

Operations Intern

Columbus, OH

- Audited manual invoicing workflows at a truck parts distributor by interviewing operations staff, identifying reconciliation bottlenecks that informed the design of an automated replacement system.
- Built an end-to-end Python automation pipeline that monitored a local folder for scanned invoices, parsed incoming documents, cross-referenced them against ERP records, and automatically emailed the manager with flagged discrepancies and the attached invoice — reducing manual reconciliation time by 70% and recapturing 15+ hours per week.

PROJECTS

HemoSave Optical Validation System | Python, C++

March – April 2024

- Fabricated a physical test rig for controlled blood-loss simulation trials and developed C++ firmware on Arduino for real-time colorimetric sensor data acquisition, with Python scripts for downstream logging and processing.
- Demonstrated proof-of-concept that light-based sensing can distinguish blood concentration, establishing the technical foundation for a <\$500 clinical device targeting 95%+ accuracy.

Robotics Kinematics Library | Python

October 2025 – Present

- Developing an open-source Python library addressing gaps in existing robotics kinematics tools, implementing Denavit-Hartenberg algorithms for multi-joint systems and balancing mathematical accuracy with accessibility for students and engineers.
- Designing an intuitive API that abstracts transformation matrices and Jacobian calculations, reducing kinematics analysis workflows from hours to minutes and lowering the barrier to entry for robotics development.

Santa/Grinch Bot | Python, OpenCV, ROS2

October – December 2025

- Designed a dual-mode autonomous robot with a multi-sensor array; developed OpenCV algorithms for real-time object classification and conducted iterative testing cycles to optimize recognition accuracy and refine mechanical and software design.

LEADERSHIP & ACTIVITIES

Black Muslim Coalition

August 2025 – Present

Secretary

- Manage internal communications and document decisions for a 58-member organization, ensuring organizational continuity and coordinating follow-through on action items across all initiatives.
- Collaborate with Triangle Region Black and Muslim student organizations to co-host cross-community events, including Black Thanksgiving and Iftar, fostering interfaith and intercultural engagement on campus.

Duke University

June 2025 – Present

Tour Guide

Durham, NC

- Lead campus tours for groups of 10–30 prospective students, tailoring content to diverse audiences including international and first-generation applicants.

Member: Theta Chi, National Society of Black Engineers (NSBE), Duke Muslim Students Association (MSA)

TECHNICAL SKILLS

Languages: Python, C/C++, Java

Tools & Frameworks: OpenCV, ROS2, MATLAB, Git, KiCad, PSpice, Oscilloscope, Function Generator, Multimeter, Power Supply, Microsoft Office

Product & Analysis: User Research, Competitive Analysis, Requirements Definition, Data Analysis, Workflow Design, Process Automation, Testing & Validation, ERP Systems

Professional: Project Management, Stakeholder Communication, Technical Documentation, Cross-functional Collaboration